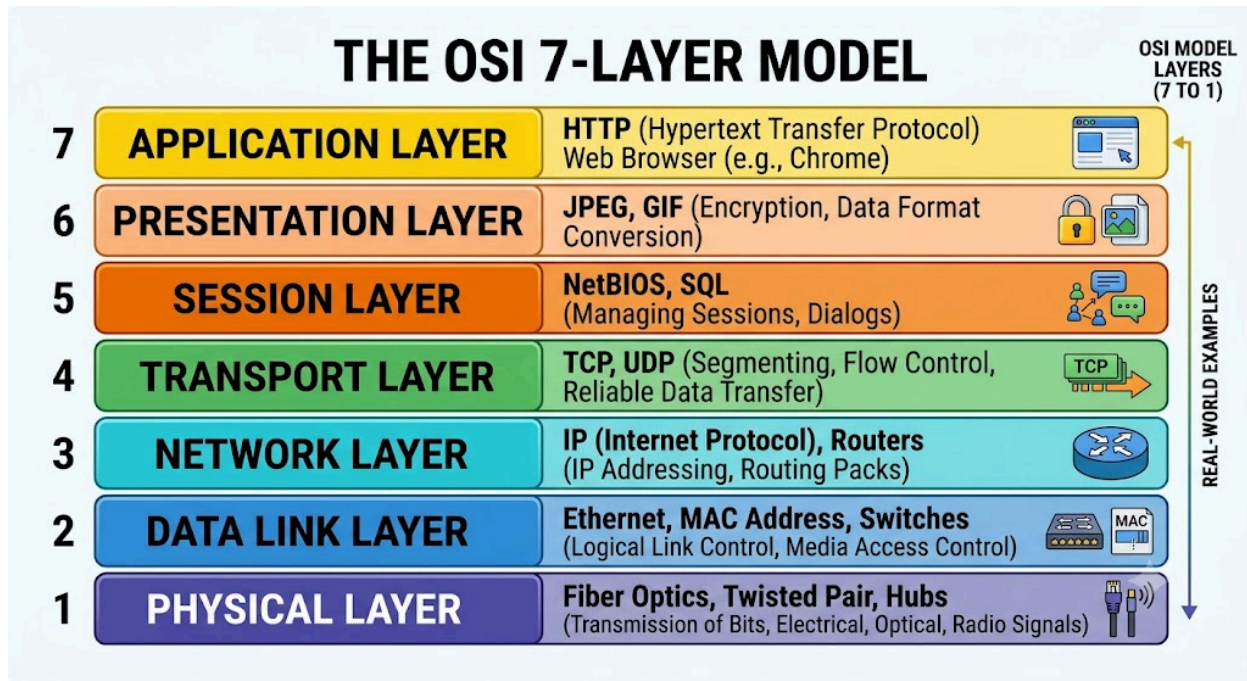


OSI LAYER

Draw a diagram of the OSI 7-layer model and label each layer with its correct name and a real-world example of a device or protocol that operates at that layer (e.g., Switch at Data Link, HTTP at Application).

ANS:



Match the following networking scenarios to the correct OSI layer: a) Encrypting data before sending, b) Routing packets across networks, c) Converting data to electrical signals, d) Opening a web page in your browser, e) Detecting and correcting errors in frames.

Hint: Write your answers as 'Scenario: Layer Name (Layer Number)'.

ANS:

- Encrypting data before sending: Presentation Layer (Layer 6)
- Routing packets across networks: Network Layer (Layer 3)
- Converting data to electrical signals: Physical Layer (Layer 1)
- Opening a web page in your browser: Application Layer (Layer 7)
- Detecting and correcting errors in frames: Data Link Layer (Layer 2)

Pick any app you use daily (like WhatsApp, Instagram, or Zomato) and describe how at least three different OSI layers are involved when you send or receive data in that app.

ANS:

Here is how three OSI layers handle sending a text message on **WhatsApp**:

Application Layer (Layer 7): WhatsApp interface captures your typed message and formats it into an application protocol request (using HTTPS/WebSockets) to begin transmission.

Transport Layer (Layer 4): Uses **TCP** to break your message into smaller segments, ensuring reliable, in-order delivery and adding port numbers (e.g., Port 443 for HTTPS) to direct traffic to the correct app process.

Network Layer (Layer 3): Encapsulates segments into IP packets and attaches the source IP (your device) and destination IP (WhatsApp server) so routers across the internet can route the message to its destination.

Given this scenario: You are unable to access a website, but you can successfully ping its IP address.

Identify which OSI layer(s) might be causing the problem and explain your reasoning.

ANS:

The problem is occurring at **Layer 7 (Application Layer)** or **Layer 6 (Presentation Layer)**.

Reasoning

- **Layers 1 through 4 are working:** Successful ICMP **ping** confirms that Physical connectivity (Layer 1), local framing/MAC addressing (Layer 2), IP routing (Layer 3), and basic host reachability (Layer 4 network pathways) are fully operational.
- **Layer 7 (Application Layer) failure:** The Web Server process (e.g., Nginx, Apache) or Web Application Firewall (WAF) running on Port 80/443 might be down, misconfigured, or dropping HTTP/HTTPS application-level requests while still answering ICMP pings.
- **Layer 6 (Presentation Layer) failure:** An expired, mismatched, or broken SSL/TLS certificate handshaking error will drop the web connection before any page content can load.
- **DNS Note:** If you are pinging the IP address directly, Layer 7 DNS resolution might also be failing if using the domain name in the browser fails.